

Flexible parametric approach to classical measurement error variance estimation without auxiliary data

Junseok Lee

August 20, 2026

1 The problem

Measurement error in a continuous covariate can introduce bias into statistical estimators. Many correction methods require information about the measurement error distribution from validation data, replicated measurements, or other auxiliary data. This paper considers the case in which such additional information is unavailable.

The goal is to estimate the unknown standard deviation v of a classical additive Gaussian measurement error using only observations of the contaminated covariate W . The main structural assumption is that the latent covariate X has compact support.

2 The setup

The classical additive measurement error model is

$$W = X + U,$$

where W is observed, X is latent, and

$$U \sim N(0, v^2), \quad X \perp\!\!\!\perp U.$$

Both the error standard deviation v and the distribution of X are unknown. The latent variable X is assumed to be continuous and compactly supported.

The support is represented by

$$X = aS + b,$$

where $a > 0$, $b \in \mathbb{R}$, and S is a continuous random variable supported on $[0, 1]$. It follows that

$$\begin{aligned} f_W(w) &= \frac{1}{v} \int_b^{a+b} f_X(x) \phi\left(\frac{w-x}{v}\right) dx \\ &= \frac{1}{av} \int_b^{a+b} f_S\left(\frac{x-b}{a}\right) \phi\left(\frac{w-x}{v}\right) dx, \end{aligned}$$

where f_S is the density of S and ϕ is the standard normal density.

Based only on

$$W_1, \dots, W_n \stackrel{\text{iid}}{\sim} W,$$

the aim is to estimate

$$(v, a, b, f_S).$$

3 The main idea

The unknown density f_S is approximated by Bernstein polynomials. For a continuous function f on $[0, 1]$, Bernstein approximation gives

$$\lim_{m \rightarrow \infty} \sup_{0 \leq s \leq 1} \left| \sum_{k=0}^m f\left(\frac{k}{m}\right) b_{k,m}(s) - f(s) \right| = 0.$$

This provides a finite-dimensional approximation to the unknown density on $[0, 1]$.

For a fixed degree m , the density of S is approximated by a mixture of Beta densities:

$$\tilde{f}_{S,m}(s; \theta_m) = \sum_{k=0}^m \theta_{k,m} \text{Beta}_{k+1, m-k+1}(s), \quad s \in [0, 1],$$

where $\text{Beta}_{\alpha, \beta}$ denotes the density of a Beta distribution with parameters α and β .

The corresponding approximation to the density of $X = aS + b$ is

$$\tilde{f}_{X,m}(x; a, b, \theta_m) = \frac{1}{a} \sum_{k=0}^m \theta_{k,m} \text{Beta}_{k+1, m-k+1}\left(\frac{x-b}{a}\right), \quad b \leq x \leq a+b.$$

Since $W = X + U$, the density of W is approximated by

$$\tilde{f}_{W,m}(w; v, a, b, \theta_m) = \frac{1}{av} \sum_{k=0}^m \theta_{k,m} \int_b^{a+b} \text{Beta}_{k+1, m-k+1}\left(\frac{x-b}{a}\right) \phi\left(\frac{w-x}{v}\right) dx.$$

The resulting log-likelihood is

$$L_n(v, a, b, \theta_m) = \sum_{i=1}^n \log \left[\frac{1}{av} \sum_{k=0}^m \theta_{k,m} \int_b^{a+b} \text{Beta}_{k+1, m-k+1}\left(\frac{x-b}{a}\right) \phi\left(\frac{W_i - x}{v}\right) dx \right].$$

For each fixed m , the parameters are estimated by

$$(\hat{v}_m, \hat{a}_m, \hat{b}_m, \hat{\theta}_m) = \arg \max_{v, a, b, \theta_m} L_n(v, a, b, \theta_m).$$

The degree m controls the flexibility of the approximation. The paper selects it using

$$\text{BIC}(m) = (m+3) \log n - 2L_n(\hat{v}_m, \hat{a}_m, \hat{b}_m, \hat{\theta}_m).$$

Thus,

$$\hat{m} = \arg \min_m \text{BIC}(m), \quad \hat{v} = \hat{v}_{\hat{m}}.$$

4 The main results

Under the Gaussian measurement error model and the compact support assumption on X , the representation of the distribution of W uniquely determines a , b , v^2 , and f_S . Therefore, the measurement error variance is identifiable from the marginal distribution of W .

For a fixed degree m , the estimator is treated as a maximum likelihood estimator under a possibly misspecified parametric model. Under regularity conditions,

$$(\hat{v}_m, \hat{a}_m, \hat{b}_m, \hat{\theta}_m) \xrightarrow{p} (v_m^*, a_m^*, b_m^*, \theta_m^*).$$

The paper also establishes

$$\sqrt{n} \left\{ (\hat{v}_m, \hat{a}_m, \hat{b}_m, \hat{\theta}_m) - (v_m^*, a_m^*, b_m^*, \theta_m^*) \right\} \xrightarrow{d} N(0, C).$$

Simulation studies show satisfactory performance for several compactly supported distributions of X . The method can face practical identifiability difficulties when the density of X has thin tails, because estimation of v is closely connected to estimation of the support of X .

5 My takeaways

- Which part of the compact support assumption makes the error variance identifiable from the distribution of W , and what difficulty arises if the support of X is unbounded?
- How sensitive is $\hat{v}$ to the BIC-selected degree $\hat{m}$, especially when the density of X is small near the endpoints of its support?

References

- [1] A. Bertrand, I. Van Keilegom, and C. Legrand. Flexible parametric approach to classical measurement error variance estimation without auxiliary data. *Biometrics*, 75(1):297–307, 2019. <https://doi.org/10.1111/biom.12960>.